

A Reproducibility-Oriented Critique of BLIP-2, LLaVA, and VisionLLM v2 with a Local-Scale BLIP-2 Reproduction

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Abstract

Multimodal large language models combine pretrained visual encoders, language models, and cross-modal alignment modules, but faithful reproduction of published systems remains difficult outside large compute environments. This paper studies three representative systems—BLIP-2, LLaVA, and VisionLLM v2—to identify where the main technical gains come from and where the principal reproducibility barriers remain. The comparative questions are threefold: which of the three systems is the most suitable target for local implementation, which design choices remain meaningful after aggressive downscaling, and how much of the observed performance gap is attributable to scale rather than to pipeline failure. To answer these questions, the study combines a structured critique of the three papers with a local reproduction of BLIP-2 using the official LAVIS codebase, reduced COCO Karpathy subsets, CLIP ViT-L, and OPT-350M on a single RTX 3070 8GB GPU. The reproduction completes stage-1 representation learning, stage-2 generative alignment, and caption fine-tuning. The best local result is obtained on a 50k/1k/1k split, reaching BLEU-4 11.13 and CIDEr 37.57, while a larger 100k run peaks early at epoch 2 and does not surpass the 50k best. The main finding is that BLIP-2 is the most defensible local reproduction target among the three papers and is reproducible at the pipeline level, but its captioning quality remains strongly constrained by backbone scale, optimization budget, and evaluation setting.

1 Introduction

Recent multimodal large language models (MLLMs) have expanded the scope of vision-language research from captioning and visual question answering to instruction-following assistants, grounding, segmentation, generation, and editing. This progress has been driven by the combination of large pretrained vision

encoders, large language models, and increasingly sophisticated alignment modules. At the same time, the field has become substantially harder to reproduce. Strong reported results are now often inseparable from large-scale pretraining corpora, multi-stage optimization, and extensive compute budgets, which makes it difficult to separate architectural contribution from scale advantage.

This tension is especially visible across three influential papers that occupy different points in the current design space. BLIP-2 introduces a modular Querying Transformer (Q-Former) that connects a frozen image encoder to a frozen language model through a compact trainable bridge (Li et al., 2023). LLaVA shifts the emphasis toward assistant behavior by aligning a vision encoder and a Vicuna-like language model through multimodal instruction tuning (Liu et al., 2023). VisionLLM v2 pushes further toward a generalist system that routes a central multimodal model into multiple task-specific decoders for perception and generation (Wu et al., 2024). In parallel, adjacent bridge-based work such as Flamingo and the original BLIP highlights a broader research trend: increasingly capable multimodal systems are being built by reusing strong unimodal backbones rather than training every component from scratch (Alayrac et al., 2022; Li et al., 2022).

The research gap addressed here is not simply whether these systems are strong, since their benchmark performance is already well established, but which aspects of their design remain reproducible and interpretable under local constraints. Existing papers report impressive empirical results, yet they do not directly answer the practical question that matters for a constrained implementation study: which model offers the best balance of novelty, openness, compute realism, and scientific interpretability when scaled down to commodity hardware? A second gap concerns pipeline fidelity. Even when official code is available, it is often un-

clear whether the full training path can be executed end-to-end on a local system without changing the method itself.

This paper addresses three questions. First, which of BLIP-2, LLaVA, and VisionLLM v2 offers the strongest local reproducibility profile? Second, once BLIP-2 is selected, can its official staged training pipeline be executed end-to-end on a single consumer GPU using principled downscaling rather than ad hoc simplification? Third, under a fixed local architecture, how does caption quality evolve as the data scale increases from 10k to 50k to 100k COCO training examples?

The paper answers these questions through a two-part methodology. The first part is a structured critique of the three papers, organized around problem formulation, core technical idea, methodology, experimental setup, results, and reproducibility implications. The second part is a local BLIP-2 reproduction based on the official LAVIS implementation, reduced COCO Karpathy subsets, CLIP ViT-L, facebook/opt-350m, and a single NVIDIA RTX 3070 8GB GPU. The implementation uses the same three-stage training logic as the original method, with targeted engineering adaptations for a local Windows environment and a subset-matched offline evaluation path for BLEU and CIDEr.

The main contributions are fourfold. First, the paper provides a reproducibility-aware critique of three representative MLLM papers rather than treating paper selection as a brief preface to implementation. Second, it documents a complete local BLIP-2 reproduction that preserves the official staged training structure. Third, it presents a controlled scaling analysis across 10k, 50k, and 100k training examples under a fixed architecture. Fourth, it shows that BLIP-2 is reproducible at the pipeline level on local hardware, but that its captioning performance remains strongly constrained by scale.

2 Comparative Critique of the Three Papers

2.1 BLIP-2

Problem and motivation. BLIP-2 addresses a central problem in multimodal learning: large language models and large vision encoders are already strong individually, but jointly training a multimodal system around them is computationally expensive and often redundant. The paper is motivated by the question of whether a lightweight trainable interface can reuse frozen unimodal back-

bones while retaining strong downstream performance (Li et al., 2023).

Key idea and approach. The core idea is the Q-Former, a compact transformer that mediates between a frozen image encoder and a frozen language model. Rather than finetuning the full multimodal stack, the method learns a small set of query-conditioned visual features that are projected into the language-model input space. This bottleneck is the paper’s main conceptual strength because it yields a precise architectural hypothesis: a carefully trained bridge may be sufficient to connect strong frozen backbones.

Methodology. The method uses two major training stages. Stage 1 trains the Q-Former against the frozen image encoder using image-text contrastive learning, image-text matching, and language modeling. Stage 2 aligns the Q-Former outputs to a frozen language model for generation. The architecture is modular, the gradient flow is interpretable, and the training objectives are clearly separated by function.

Dataset and experimental setup. The full paper uses a 129M image-text pretraining mixture assembled from COCO, Visual Genome, CC3M, CC12M, SBU, and LAION subsets, followed by downstream evaluation and finetuning on established benchmarks (Li et al., 2023). The strongest reported configurations use large frozen backbones such as ViT-g and OPT-2.7B, with training performed on a 16-A100(40GB) machine for multiple days.

Results. BLIP-2 reports strong zero-shot and finetuned results across VQA, captioning, and retrieval. A particularly striking result is that it surpasses Flamingo-80B on zero-shot VQAv2 by 8.7 absolute points while using $54\times$ fewer trainable parameters (Li et al., 2023). On COCO captioning, the reported ViT-g OPT-2.7B configuration reaches BLEU-4 43.7 and CIDEr 145.8.

Main takeaways. BLIP-2 is the strongest local reproduction target among the three papers because its method remains scientifically coherent after downscaling. However, the paper’s “compute-efficient” framing is relative to prior frontier systems, not to commodity hardware. The method is modular and reproducible in principle, but paper-level performance remains tightly coupled to large data and backbone scale.

2.2 LLaVA

Problem and motivation. LLaVA addresses a different problem: multimodal systems may recognize images, yet still behave poorly as assistants because they are not trained on conversational and instruction-following supervision. The paper is motivated by the need for a practical, open visual assistant rather than by a new bridge architecture alone (Liu et al., 2023).

Key idea and approach. The central idea is simple and influential: a frozen CLIP image encoder can be linked to a Vicuna-style language model through a lightweight projection layer, after which the combined model can be trained on multimodal instruction-following data. The paper’s main contribution is therefore data-centric rather than deeply architectural.

Methodology. LLaVA uses a two-step procedure: feature alignment between the vision encoder and the language model, followed by visual instruction tuning. The instruction data are generated by GPT-4 from symbolic image descriptions, and the training set contains 158K multimodal instruction-response pairs (Liu et al., 2023). The design is methodologically appealing because it demonstrates how much assistant-like behavior can be induced by supervision format.

Dataset and experimental setup. The paper uses CLIP ViT-L as the visual backbone and Vicuna as the language model, with training conducted on 8 A100 GPUs. The reported instruction-tuning phase completes in roughly 10 hours (Liu et al., 2023), which is considerably more accessible than the largest BLIP-2 pretraining setup, although still well beyond a single 8GB consumer GPU.

Results. LLaVA demonstrates strong open-ended conversational behavior and reports state-of-the-art performance on ScienceQA when combined with GPT-4 as a judge, reaching 92.53% (Liu et al., 2023). The paper’s impact is also visible historically: it established visual instruction tuning as a dominant pattern in open-source multimodal assistant research.

Main takeaways. LLaVA is compelling because it reorients the problem from representation learning to alignment and supervision. Its main limitation, from a reproduction perspective, is that a large part of its success is entangled with the GPT-generated instruction pipeline. The architecture

is comparatively simple, but faithful reproduction depends on recreating the data-generation strategy.

2.3 VisionLLM v2

Problem and motivation. VisionLLM v2 targets the broadest problem of the three papers: most multimodal systems remain narrow, producing primarily text outputs while relying on separate specialist models for detection, segmentation, pose estimation, and generation. The paper is motivated by the goal of a single end-to-end generalist system for many vision and vision-language tasks (Wu et al., 2024).

Key idea and approach. Its central mechanism is the “super link,” which combines routing tokens and task-specific learnable queries to connect a core multimodal language model to multiple external decoders. This design enables end-to-end gradient flow from task-specific objectives back into the shared model, which is more ambitious than a tool-use style interface.

Methodology. The methodology is multi-stage and multi-decoder. The system integrates decoders such as Grounding DINO, UniPose, Stable Diffusion v1.5, and InstructPix2Pix, and trains the overall model across a very broad task set (Wu et al., 2024). The resulting system is impressive, but methodologically harder to isolate because gains may derive from routing, decoder reuse, data scale, or task breadth.

Dataset and experimental setup. The experimental setup is the most demanding of the three papers. The appendix reports 64 A100(80GB) GPUs for stage 1 and 128 A100 GPUs for later stages, with total training time on the order of 18 days (Wu et al., 2024). The system also depends on multiple pretrained decoders and a large collection of datasets and task templates.

Results. VisionLLM v2 reports broad generalist capability and competitive specialist performance. For example, it reports 56.7 AP_b on COCO object detection with a Swin-T backbone, competitive with Grounding-DINO-T, along with strong results in pose and segmentation settings (Wu et al., 2024). Its contribution is therefore breadth more than a single benchmark win.

Main takeaways. VisionLLM v2 is an important systems contribution, but it is not a viable local reproduction target for this study. Its engineering

Paper	Problem focus	Central idea	Experimental scale	Local reproduction verdict
BLIP-2	Connect frozen vision and language backbones efficiently for multimodal generation and understanding	Two-stage Q-Former bridge between a frozen image encoder and a frozen LLM	129M image-text pairs; multi-day training on 16 A100 GPUs for large variants	Best target: modular, open, and interpretable after downscaling
LLaVA	Build a practical multimodal assistant rather than only a captioning or representation model	Lightweight vision-language projection plus multimodal instruction tuning	8 A100 GPUs; GPT-generated instruction data; moderate but still non-local compute	Viable only with stronger hardware and the instruction-data pipeline
VisionLLM v2	Unify many vision and vision-language tasks inside one end-to-end system	Super-link routing between a central MLLM and multiple specialist decoders	64–128 A100 GPUs; multi-stage training across many tasks and decoders	Not viable for a local reproduction study

Table 1: Comparative summary of the three papers from a reproducibility-oriented perspective.

complexity and compute requirements are too high for local implementation, and its scientific contribution is more difficult to disentangle cleanly under heavy downscaling.

3 Local BLIP-2 Reproduction

The reproduction focuses on technical execution rather than on reiterating the model selection argument. The objective is to preserve the official staged BLIP-2 training logic while aggressively reducing scale so that the full pipeline can run on a single RTX 3070 8GB GPU. Success is therefore defined at the pipeline level: valid completion of stage 1, stage 2, and caption finetuning, together with reproducible evaluation on a fixed local benchmark. The completed runs should be read as successive attempts within one local reproduction, not as separate environments: a 10k pilot attempt, a 50k main attempt with a short 50k refinement rerun, and a 100k long-run attempt.

3.1 Pipeline and Mathematical Formulation

BLIP-2 consists of three components: a frozen vision encoder E_v , a trainable Q-Former Q_θ , and a frozen language model E_l (Li et al., 2023). Given an image x , the vision encoder produces visual features $V = E_v(x)$. A fixed set of learned query tokens interacts with V through the Q-Former to

produce query-conditioned outputs

$$Z = Q_\theta(V).$$

These outputs are projected into the language-model input space through a learned projection matrix W , producing the soft visual prompts that condition generation.

Stage 1 trains the bridge against the frozen image encoder using the same local LAVIS objective family as the official implementation:

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{itc}} + \mathcal{L}_{\text{itm}} + \mathcal{L}_{\text{lm}}.$$

Here, $\mathcal{L}_{\text{itc}}$ enforces image-text contrastive alignment, $\mathcal{L}_{\text{itm}}$ trains a matching classifier over fused representations, and $\mathcal{L}_{\text{lm}}$ encourages the bridge to support caption-token prediction.

Stage 2 aligns the bridge to the frozen language model through an autoregressive objective:

$$\mathcal{L}_{\text{stage2}} = - \sum_{t=1}^T \log p(y_t \mid y_{<t}, WZ),$$

where y denotes the target caption tokens. Caption finetuning uses the same image-conditioned generative path but specializes it to COCO captioning with the prompt a photo of. The local configuration keeps 32 query tokens throughout.

3.2 Local Data, Configuration, and Evaluation Protocol

The local benchmark is derived from deterministic subsets of the COCO Karpathy caption splits (Lin et al., 2014). Three training scales are used: 10k/1k/1k, 50k/1k/1k, and 100k/1k/1k for train/validation/test. Validation and test remain fixed at 1,000 images across all runs so that internal comparisons reflect model behavior rather than split changes.

The local model uses CLIP ViT-L as the visual backbone, OPT-350M as the language model, image resolution 224, 32 query tokens, automatic mixed precision, and gradient accumulation of 8. The 50k attempt family uses a 3/3/5 schedule for stages 1, 2, and caption finetuning, while the 100k long-run attempt uses a 10/10/15 schedule. Evaluation is performed offline because the stock Windows METEOR path was unstable in the local environment. The checkpoint-producing caption runs therefore use report_metric=false, after which BLEU and CIDEr are computed from saved

val_epoch*.json files against subset-matched COCO-format ground truth.

Dimension	BLIP-2 paper configuration	Local reproduction
Pretraining data	129M image-text pairs from COCO, Visual Genome, CC3M, CC12M, SBU, and LAION subsets (Li et al., 2023)	Reduced COCO Karpathy subsets only: 10k, 50k, and 100k training examples
Bridge schedule	250k steps in stage 1 and 80k steps in stage 2 (Li et al., 2023)	3 epochs / 3 epochs in the 50k attempts; 10 epochs / 10 epochs in the 100k long run
Caption setup	5 epochs, batch size 256, image resolution 364 (Li et al., 2023)	5 caption epochs in the 50k attempts or 15 in the 100k long run, batch size 1–2 with accumulation 8, image resolution 224
Vision encoder	ViT-g/14 for the strongest OPT caption result (Li et al., 2023)	CLIP ViT-L
Language model	OPT-2.7B (Li et al., 2023)	OPT-350M
Compute	Multi-day training on 16 A100 GPUs for large variants (Li et al., 2023)	One RTX 3070 8GB GPU; multi-hour to multi-day desktop runs

Table 2: Original BLIP-2 scale and the local reproduction.

3.3 Implementation Adaptations

The reproduction retains the official LAVIS code path, but several targeted engineering adaptations are required to make it reliable in a local Windows setting. First, non-distributed guards are added around collective operations and barriers so that the training loop functions in single-process mode. Second, the local OPT-350M path requires a projection fix because the Q-Former outputs must be mapped into the actual token-embedding dimension rather than the hidden-size assumption used for larger OPT variants. Third, local cache-path and image_id fixes are required so that the reduced COCO subsets align correctly with the evaluator. Fourth, the metric path is reconstructed offline with subset-matched ground truth because Windows METEOR was unstable. Fifth, per-epoch checkpoint saving and resume hooks are added so that long runs remain recoverable and every caption epoch can be scored after training.

3.4 Results and Comparison

Across the completed local attempts, all three stages run successfully on the 10k pilot, the 50k

line, and the 100k long-run configuration. The 50k line includes a main attempt and a short refinement rerun; the strongest completed local caption result is the main attempt selected at epoch 3. This result reaches BLEU-4 11.13 and CIDEr 37.57 on the paper-style $\times 100$ scale. The 100k long-run attempt is operationally successful and more diverse, but its best checkpoint occurs at epoch 2 and still underperforms the 50k best.

Metric	Paper	Local best
BLEU-4	43.7	11.13
CIDEr	145.8	37.57

Table 3: BLIP-2 paper result and the best local completed result. The values are not directly leaderboard-equivalent because the local system uses smaller backbones, less data, lower resolution, and a reduced validation split.

Run	Scale	BLEU-4	CIDEr	Unique captions
10k pilot (final e0)	10k	1.45	3.03	61
50k main (best e3)	50k	11.13	37.57	563
50k refinement (final e4)	50k	10.49	37.14	607
100k long-run (best e2)	100k	8.57	25.34	677
100k long-run (final e14)	100k	6.61	18.36	734

Table 4: Internal scaling comparison across completed local runs.

The local scaling trend is therefore clear. Increasing the training set from 10k to 50k produces a large quality gain, which indicates that the local pipeline is learning meaningful cross-modal structure rather than emitting arbitrary captions. The transition from 50k to 100k is different: caption diversity continues to rise, but BLEU and CIDEr peak early and then decline.

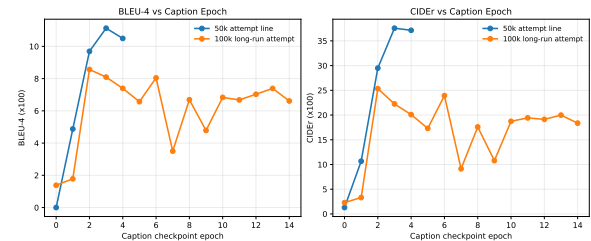


Figure 1: Caption metric trajectories for the completed 50k and 100k paths. The 100k run peaks early and then degrades, so best-checkpoint selection by validation CIDEr is empirically necessary rather than optional.

The training losses support the claim that the pipeline itself is functioning. The 50k main attempt ends stage 1 at loss 0.412 and stage 2 at

0.767, while the 100k long-run attempt ends stage 1 at 0.330 and stage 2 at 0.363. Caption loss also decreases steadily in the long run. The main failure mode therefore appears in model selection and final caption quality rather than in the basic operability of the training stages.

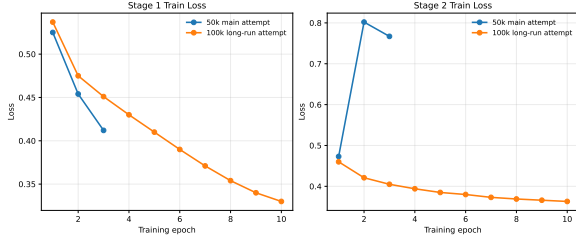


Figure 2: Stage-1 and stage-2 train-loss trajectories for the 50k and 100k runs. The smooth loss decrease in the 100k run indicates that later caption degradation is better interpreted as a selection or overtraining issue than as a bridge-stage instability.

100k checkpoint	BLEU-4	CIDEr	Unique captions
Best epoch 2	8.57	25.34	677
Final epoch 14	6.61	18.36	734

Table 5: Best and final checkpoints for the 100k long run.

Under the fixed local architecture used here, larger data and a much longer schedule do not automatically yield better captioning quality. The evidence points to an optimization and checkpoint-selection problem rather than a failure of the staged BLIP-2 design itself.

4 Limitations and Validity

The conclusions are informative but bounded by several limitations. The local evaluation uses a fixed 1,000-image validation subset rather than the full Karpathy COCO test set, so the reported numbers are directly comparable across local runs but not directly comparable to the paper’s leaderboard setting. The local architecture also differs materially from the strongest BLIP-2 captioning configuration: CLIP ViT-L and OPT-350M replace ViT-g and OPT-2.7B, and the image resolution is reduced from 364 to 224.

A second limitation concerns systems validity. The reproduction relies on targeted code adaptations to make the official LAVIS path work in a single-process Windows environment. These modifications do not change the intended training logic, but they do mean that the study measures both

model reproducibility and the portability of a research codebase into a non-native environment. A third limitation is that all completed runs are single-seed trajectories. The ranking between the 50k and 100k runs is therefore empirically grounded but not variance-estimated.

5 Discussion

One useful distinction preserved from the larger draft is between pipeline reproducibility and performance reproducibility. At the pipeline level, the answer is positive: after targeted engineering fixes, the official BLIP-2 path can run end to end on a single consumer GPU, produce recoverable checkpoints, and support stable offline BLEU/CIDEr evaluation. At the performance level, the answer is more limited. The local system remains far below the paper’s captioning quality even after the clear improvement from the 10k pilot to the 50k main attempt. Treating these as separate outcomes makes the reproduction result more precise and more scientifically useful.

The project also contributes an artifact trail rather than only a final score. The reported metrics are traceable to saved validation predictions and subset-matched ground-truth files, and the later 50k and 100k attempts are continuations of the same codebase rather than disconnected experiments. This matters because the 100k result shows that schedule design and checkpoint selection matter as much as raw runtime: the pipeline stays stable while caption quality peaks early. The work is therefore useful not only as a numerical comparison, but also as a reusable local BLIP-2 reproduction scaffold.

6 Conclusion

This paper evaluated BLIP-2, LLaVA, and VisionLLM v2 through a reproducibility-oriented critique and then implemented a local-scale BLIP-2 reproduction on a single RTX 3070 8GB GPU. The comparative analysis shows that BLIP-2 is the most suitable target for a local implementation study because its Q-Former architecture remains interpretable after downscaling, whereas LLaVA depends more heavily on a synthetic instruction-data pipeline and VisionLLM v2 depends on very large compute and decoder breadth.

The reproduction confirms that the BLIP-2 pipeline is executable end-to-end at local scale. Stage-1 representation learning, stage-2 generative

alignment, and caption fine-tuning all complete successfully, and the local evaluation path produces stable BLEU/CIDEr estimates from saved checkpoints. The best local result, obtained on the 50k main attempt, reaches BLEU-4 11.13 and CIDEr 37.57; the larger 100k long-run attempt also completes successfully, but peaks early and does not surpass the 50k best.

The central conclusion is straightforward. BLIP-2 is reproducible at the pipeline level on commodity hardware after targeted engineering work, but its published captioning quality is not reproducible at that scale. The remaining performance gap is best explained by backbone size, data scale, optimization budget, and evaluation setting rather than by a failure of the staged training method itself.

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